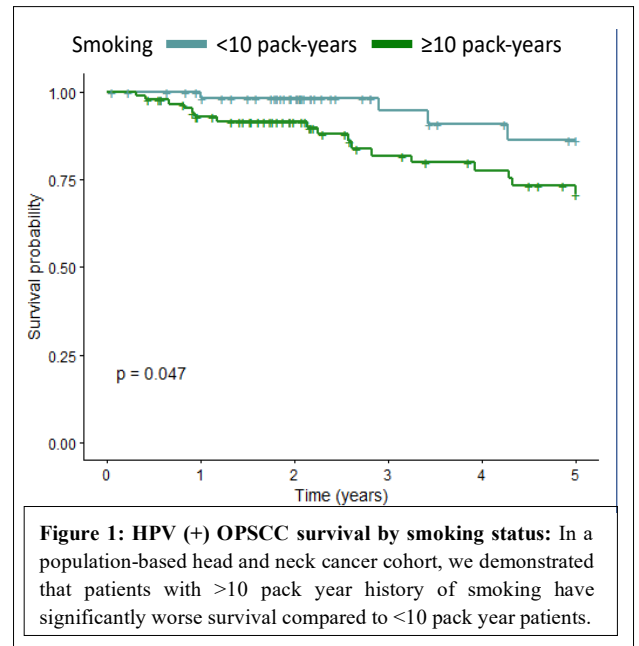


BACKGROUND

Oropharyngeal squamous cell carcinoma (OPC) is among the most common cancers of the upper aerodigestive tract and includes cancers arising from the tonsils, base of tongue, soft palate, and lateral and posterior pharyngeal walls. Approximately 60-70% of new OPC cases are now attributable to human papillomavirus (HPV).¹ HPV (+) OPC incidence has continuously risen since the 1980s in the United States;² in the last 10 years HPV (+) OPC has overtaken cervical cancer as the most common HPV-related malignancy in the United States.³ Oral oncogenic infection is a necessary but insufficient precursor to HPV (+) OPC.⁴ Previous studies have found that among unvaccinated men aged 18 to 59, the prevalence of oncogenic and non-oncogenic oral HPV infection was 1.6% and 8.4%, respectively.⁵ While the vast majority of people clear the infection, a subset of individuals develop chronic HPV infection that can lead to HPV (+) OPC.^{6,7}



HPV (+) OPC has been shown to respond more effectively to radiation and chemotherapy compared to HPV (-) head and neck cancers.⁸⁻¹⁰ Therefore, current treatment paradigms and clinical trials for OPC have focused on de-intensification of therapy and minimally invasive surgical approaches to limit the morbidity of treatment and improve functional outcomes.¹¹⁻¹⁴ However, there is evidence from our group and others that OPC in Veterans is associated with worse oncologic outcomes, and therefore widespread adoption of **de-escalation may in fact be unsafe**. Veterans have significantly higher smoking rates compared to the Veteran population, and tobacco exposure generates a complex interaction with HPV in the context of OPC. First, as a known carcinogen, tobacco itself leads to OPC development.^{15,16} Second, tobacco use may independently facilitate the oncogenic effects of HPV. Heavy smokers have nearly four times the prevalence of oral HPV compared with non-smokers.¹⁷ Our group and others have now reported data from multiple, non overlapping, prospective and retrospective cohorts, demonstrating conclusively that HPV (+) OPC in current or former smokers is associated with substantially inferior, cancer specific survival.^{1,18}

The Veterans Health Administration (VHA) is the largest integrated healthcare system in the United States.¹⁹ Veterans are significantly older²⁰ and have greater exposure to tobacco^{21,22} compared to the general population. The absence of extant estimates of OPC burden in the VHA makes it impossible to develop optimal oncologic clinical practices and to develop OPC oriented clinical trials for this understudied patient population, thus creating a critical clinical gap. As the incidence of HPV(+) OPC continues to increase, there is evidence that the Veteran population will be disproportionately affected. Our published analysis of SEER and VA CDW data indicates that the increase in OPC incidence in the Veteran population may in fact be **3 times greater than that at the national level**.²³ Even while life-saving methods of primary (HPV vaccination) and secondary (smoking cessation) prevention methods are available, there are currently no tools in place to understand the current and future burden of OPC among Veterans to guide intervention target and set up policy prioritization. **This represents a critical population health gap.**

It is now well established that HPV vaccination is an effective method of primary prevention of cervical and HPV(+) OPC. However, we have recently demonstrated that eligible Veterans have significantly lower rates of HPV vaccination than their civilian counterparts (**Figure XX**). This alarming finding raises the possibility that rates of HPV+ OPSCC will remain higher for decades longer within the VHA, resulting in increased morbidity

and mortality for Veterans and increased cost for the VHA. In this proposal, we seek to understand the current and future burden of HPV (+) OPC in the Veteran population by examining current outcomes in HPV(+) OPC among Veterans, evaluating risk factors for under-vaccination, and modeling the future impact of HPV(+) OPC on Veteran survival and quality of life. Completion of the Aims detailed below will generate a clinically meaningful impact on individual Veterans with OPC and potentially generate more cost-effective approaches to prevent and control this deadly disease by providing actionable pathways to saving lives through primary and secondary prevention within the VHA. **We hypothesize that current vaccination trends portend a continued increase in OPC among the Veteran population through the next four decades that requires understanding and quantification. Furthermore, we hypothesize that understanding the future burden of OPC in the VHA can lead to the design of targeted interventions around HPV vaccination and smoking cessation that account for demographic and geographic risk factors and barriers to care.**

SIGNIFICANCE

This proposal utilizes unique national VHA databases to advance our understanding of HPV (+) OPC epidemiology and outcomes in the US Veteran population, and will identify strategies to reduce the burden of this disease in a population that is at greater risk poor outcomes. Our proposed study will reveal differences in HPV-vaccination rates at the VHA by a number of demographic and regional variables, allowing for the identification of at-risk Veteran populations. Understanding differences in vaccination rates across the US and barriers to vaccination at the VHA creates the potential for targeted intervention strategies. It is important to note that African American Veterans with OPC can expect equivalent treatment and achieve equivalent survival within VHA institutions as we have previously shown.²⁴ However, a disparate OPC impact can obviously be generated by a disproportionate under-vaccination policy, as well as other minority populations. In FY 2019, 16.9% of active duty enlisted men and 28.9% of active duty enlisted women were African American. **The potential for a disparate impact arising from non-targeted and non-data driven vaccination policies is therefore substantial.**

By developing VISN-specific data, we will provide VHA leadership and key stakeholders with the tools to implement the most effective targeted vaccination strategies unique to different regions. This proposal directly addresses the HSR&D Research priorities of “Population Health and Whole Health” as it focuses heavily on understanding and increasing HPV-vaccination uptake at the VHA, a proven method of primary prevention for HPV-associated malignancies. Moreover, our proposal aligns with ORD-wide Research Priorities as it will take advantage of existing administrative datasets at the VHA (“optimize VA research enterprise”) to help understand OPC disease burden, propose targeted interventions such as vaccination and smoking cessation initiatives (“putting VA data to work for veterans”), and model the impact of these interventions (increase substantial real-world impact of VA research). While this study is focused on improving Veteran health, the results will have far-reaching implications for the entire military community. The data can be used to measure HPV vaccination uptake and inform primary prevention strategies in active duty military personnel. Our algorithms could also be adapted to active duty military databases in order to capture at-risk military populations earlier.

INNOVATION AND IMPACT

To the best of our knowledge, this is the first study to examine barriers to HPV vaccination among Veterans, and the first to project the impact of primary prevention on future OPC incidence and outcomes. While these factors have been studied in civilian populations, we believe that extrapolating general population data to the Veteran population is inappropriate for two reasons. First,

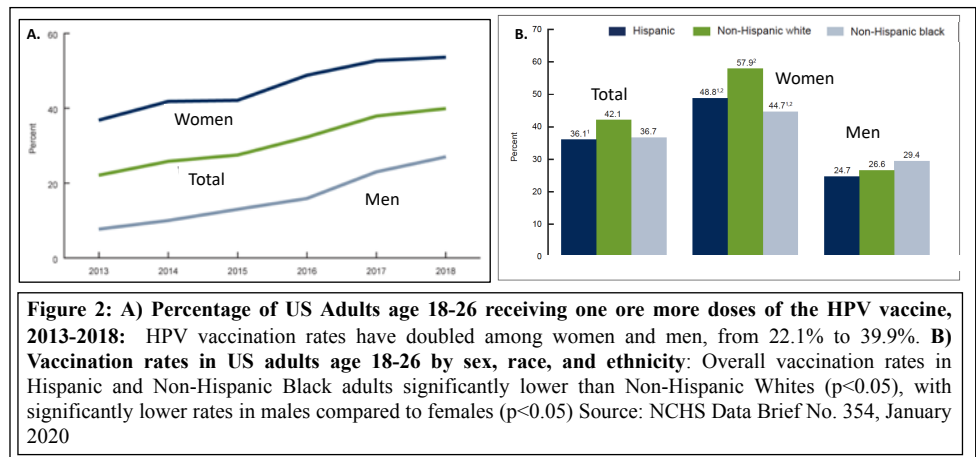
HPV vaccination rates in active duty military and Veteran populations are extremely low,²⁵ in contrast to civilian populations, where rates of HPV vaccination doubled between 2013 and 2018 (**Figure 2**).²⁶ The Center for Disease Control recently expanded HPV vaccination indications to include men and women <45 years of age making many Veterans entering the VHA system eligible.²⁷ There is currently no requirement for HPV vaccination among active-duty military or Veterans,^{25,28} yet recent data demonstrated that 19% of males in active duty developed a new HPV16 or HPV18 oncogenic infection over ten years of service.²⁹ Second, Veterans smoke at a higher rate compared to the general population^{21,22} and the 2019 Survey of Veteran Enrollees' Health and Use of Health Care showed no change in the rate of smoking as well as a high rate of unsuccessful cessation attempts.³⁰

Over the last decade, there has been an evolution in eligibility criteria for HPV vaccination. CDC guidance on HPV vaccination currently supports vaccination for all adults “through age 26 years if not adequately vaccinated previously”.³¹ This guidance supports a clear line of potential intervention whereby active service members not previously fully immunized can receive “catch-up” vaccination upon entry into service. Furthermore, more recent CDC guidance provides that adults 27-45 may receive the HPV vaccine if deemed appropriate by a physician; this guidance would therefore cover a majority of active duty members and extend substantially into the post-active duty, Veteran population as well. Coupled with targeted efforts at smoking cessation, we believe that, a data-informed strategy of HPV vaccination within the active duty and Veteran population would dramatically blunt the expected mortality and morbidity of HPV(+) OPC generating a broad and significant impact across the VHA **for at least several decades**.

PATIENT EXPERIENCE AND VETERAN ENGAGEMENT

Veteran input will be essential to the overall success of this proposal. Once we understand the barriers to vaccination from the population perspective, it will be imperative to seek the input of Veterans on their understanding of HPV vaccination, the availability of HPV vaccination, and the link between HPV and oropharyngeal cancer. To that end, we will engage with key Veteran groups through an established VHA Research Week at the Michael E. DeBakey Houston VA Medical Center. Led by co-investigator Vlad Sandulache, MD, PhD....

The results of this study will set the foundation for critical qualitative research on barriers to HPV vaccination among Veteran populations. At the conclusion of Aim 1, we will for the first time have a complete understanding of HPV vaccination prevalence among Veterans, as well as populations at risk for undervaccination. In order to understand barrier to vaccination, direct input from eligible veterans will be essential. We will work with VISNs to develop Veteran focus groups and interviews to understand specific barriers to vaccination. At the conclusion



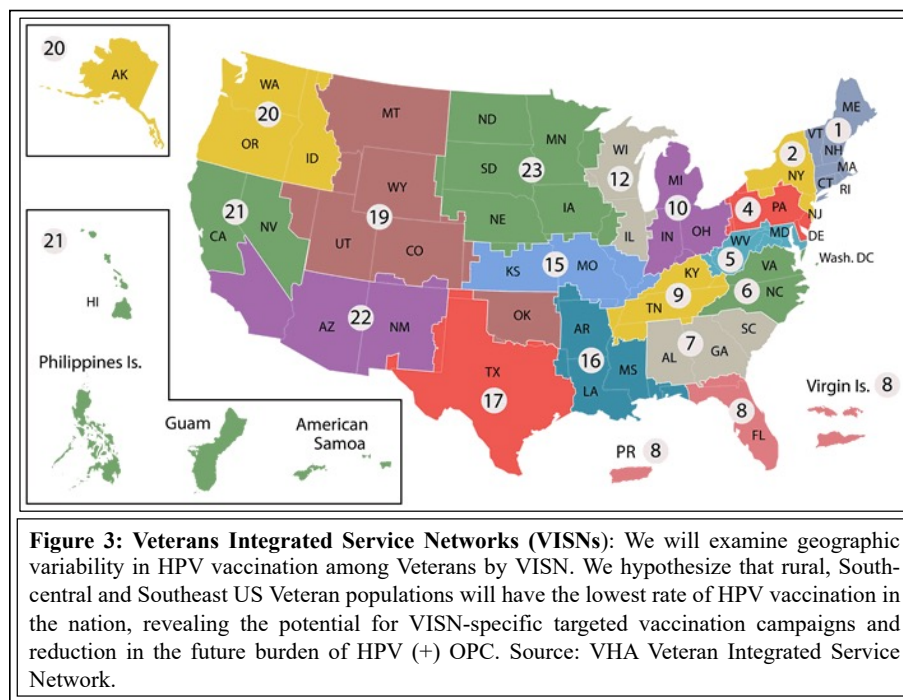
of Aim 2, we will understand oncologic outcomes in Veterans with OPC. We will seek input from Veteran regarding attitudes on treatment deintensification. This will be essential to design the best and most appropriate HPV(+) OPC clinical trials that are tailored to the specific needs of the VHA population. Aim 3 will result in novel projections of future HPV(+) OPC burden in the VHA population that could be used to design primary (vaccination) and secondary (smoking cessation) interventions. We will seek Veterans perspectives on these data, and seek Veteran input on the most effective ways to disseminate our results across the VHA.

OVERLAP

There are currently no studies being conducted at the VHA or across other institutions that seek to examine HPV vaccination uptake in the Veteran population. The majority of these studies in the civilian population focus on sociodemographic groups at risk of under-vaccination with an emphasis on adolescents. We identified one study (NCI funded study, 5 P30CA014520-47) at the University of Wisconsin-Madison titled, “Community Outreach and Engagement” that broadly seeks to decrease cancer incidence and mortality among multiple groups in the community, including Veterans. They specifically mention improvement of HPV vaccination uptake in the community but do not focus primarily on the Veteran population. Additionally, they are not focused on HPV(+) OPC.

Table 1. Research Overlap	HSR&D	QUERI	Clinical Trials	RePORTER
HPV+Vaccination + Veterans Affairs	0	0	0	1
HPV-Vaccination	0	0	482	203
HPV + Oropharyngeal Cancer	0	0	112	703
HPV-Vaccination + Oropharyngeal Cancer	0	0	12	45

RESEARCH DESIGN AND METHODS



Aim 1. Define the prevalence of HPV vaccination among eligible Veterans and identify populations at highest risk for undervaccination: Using data from the VHA Corporate Data Warehouse (CDW), we will estimate the prevalence of HPV vaccination in Veterans <45 years of age and examine vaccination rates by race/ethnicity, socioeconomic status, smoking status, rurality based on Rural-Urban Commuting Area Codes (RUCAs), and geographical location based on Veteran Integrated Service Network (VISN) (Figure 3). We will include all Veterans with at least one primary care visit from 2018 to 2023 aged <45 years. VISNs are responsible for VHA healthcare planning and resource allocation in a particular

geographic region,³² and thus present an ideal opportunity to assess geographic variability in vaccination rates and ultimately apply targeted interventions. We hypothesize that smokers and African-Americans have the lowest rate of HPV-vaccination, and HPV vaccination rates vary significantly across regions, with rural, South-central, and Southeast regions demonstrating the lowest rates of HPV vaccination. This approach will identify at-risk groups based on demographic or regional characteristics and allow for tailored VHA or VISN-specific HPV vaccination campaigns or other cancer prevention strategies.

Experimental Approach.

Study Population: We will construct a cross-sectional, national sample of Veterans to calculate the yearly prevalence of HPV vaccination with data from the VHA CDW. To ensure that we capture a sample that qualifies for HPV vaccination and engages with the VHA and military health care system, we will include all Veterans with at least one primary care visit from 2018 to 2023 aged <45 years. In 2017, 6.1 million Veterans had one VA visit, of which 1.7 million were aged <45 years.³³

Experiment 1.1 Prevalence of HPV Vaccination among eligible Veterans: We will first abstract HPV vaccination records, from eligible Veterans <45 years of age as well as the date of each HPV vaccination dose. HPV vaccination recommendations include a series of 3 doses at minimum intervals of 4 weeks between the first and second dose, 12 weeks between the second and third dose, and five months between the first and third dose.³¹ Therefore, we will classify HPV vaccination into multiple categories that account for the initiation of vaccination and compliance as outcome variables: (1) any HPV vaccination (yes/no), (2) number of HPV vaccines (0-3), (3) completion of 3-dose HPV vaccination recommendations (yes/no), and (4) HPV vaccine time as routine (yes/no) if vaccinated between age 9 through 12 years, catch-up as 13 to 26 years, and shared-decision making as vaccinated between ages 26 to 45 years.

Experiment 1.2 Race/ethnicity, rurality, geographical location, and socioeconomic status as potential risk factors for under-vaccination: We will then evaluate associations between multiple demographic factors and the likelihood of HPV vaccination among eligible Veterans. Race/ethnicity will be obtained from the VistA Patient (#2) file in MedSAS and will be categorized into non-Hispanic white, non-Hispanic black, Hispanic, and other based on the algorithm created by VA Information Resource Center.³⁴ Individual zip codes will be obtained in MedSAS and will be mapped to county-level data to calculate county-level estimates of socioeconomic status using a geospatial score from the 2010 US Census Bureau American Community Survey that addressed education, employment and occupation, housing conditions, income and poverty, racial composition, and residential stability.³⁵ Individual zip codes will also be linked to RUCAs. RUCAs were developed by the US Department of Agriculture to delineate rural-urban census tracts, and have been adopted by the VHA Office of Rural Health as the primary stratification system for defining rural/urban status.³⁶ RUCAs have advantages over other rural-urban measures since they take into account not only population density but also a functional measurement of travel distance. We will classify RUCAs into four categories, urban, large rural, small rural, and isolated. Other risk factors, such as age on the index date will be obtained from MedSAS. Smoking status will be derived from automated clinical reminders known as Health Factors in VHA CDW.^{37,38} The VHA requires providers to complete these Health Factors regularly within the electronic health records. The results from these structured EHR data are housed in the CDW and refreshed daily. Smoking status will be determined using a VHA-specific validated structured data tool.^{37,38} We will classify smoking as former, current, and never on the index date (i.e., first primary care visit within the study period) or within one year before the index date.

Statistical Analysis. *Unadjusted analyses:* We will examine the prevalence of HPV vaccination compliance, completion, and time frame across the entire cohort, as well as differences in the prevalence or completion of HPV vaccination by smoking status, race/ethnicity, VISN, or rural-urban context. Chi-square tests will be used

to determine if there are differences in the prevalence of HPV vaccination between strata of smoking status, race/ethnicity, VISN, or rural-urban context. We will also calculate prevalence differences to measure the potential public health impact of the risk factor by calculating the number of additional Veterans vaccinated. Adjusted analyses: We will use logistic regression for outcomes (1) modeling no (vs any) HPV vaccination; (2) modeling no completion (vs completion) of HPV vaccination recommendations; and (3) modeling no routine (vs routine) HPV vaccination (see definition in *HPV Vaccination* section). For outcome (2), we will use multinomial logistic regression for HPV dose (0, 1 or 2, and 3). To identify barriers to HPV vaccination, we will include all of the aforementioned candidate risk factors as independent variables in the regression analyses. If any geospatial differences in the prevalence of HPV vaccination are identified in the unadjusted analyses, we will consider the innate relationships of individuals within a neighborhood with a multilevel logistic regression or multilevel multinomial logistic regression, as appropriate.³⁹ Additionally, we will study the interaction between these risk factors and VISNs to determine the best methods for targeting intervention.

Expected results, pitfalls and alternative approaches. We expect that HPV vaccination is lower among African-American Veterans compared to non-Hispanic Whites, as has been noted in the civilian population (**Figure 2**).²⁶ The VHA Office of Rural Health has determined that Veterans living in rural communities often lack adequate access to basic healthcare and preventative care,⁴⁰ and we hypothesize that HPV vaccination rates will be lower among Veterans living in rural communities. We also expect Veterans in south-central and southeast VISNs to have the lowest rate of HPV vaccination. This hypothesis is derived from civilian HPV vaccination data among adolescents, where these regions have among the lowest rates of vaccination in the nation.^{41,42} While data from civilian populations serve as hypothesis-generating, the lack of data on barriers to vaccination within the VHA severely limits the potential for tailored prevention programs. Aim 1 is designed to address this critical research gap. With at least 1.7 million samples (and including 283,000 females) in the VHA <45 years of age, we will be adequately powered to detect any association. However, with this large number of samples, there is the risk that every test will be statistically significant, especially when comparing the prevalence of HPV vaccination across variables. Thus, we will focus *a priori* on effect sizes and differences in prevalence rather than p-value.

Aim 2. To measure the impact of tobacco exposure on HPV(+) OPC outcomes in a contemporary VHA cohort.

Our preliminary data indicate that Veterans present with: 1) a higher OPC incidence, 2) a disproportionate rate of high-risk OPC associated with smoking, 3) decreased progression free survival (institutional data) and 4) decreased overall survival (VHA Corporate Data Warehouse (CDW) data) compared to the non-Veteran population.⁴³⁻⁴⁵ This generates a clear and present clinical problem which must be aggressively tackled over the coming decade. However, the lack of high-quality, multi-institutional data precludes development of robust, prospective clinical trials targeting OPC in Veterans. In this Aim, we will bridge the gap between high-quality, granular institutional datasets over the last decade and the less granular data available through the VA CDW. Using a VHA OPC working group established at the 2019 OPC Field Based Meeting we will assemble and analyze a multi-institutional Veteran cohort of ~1,500 Veterans with HPV (+) OPC (2010-2018) to determine the relative impact of smoking on progression free survival (PFS) and overall survival (OS) and to compare and contrast these effects with those generated in non-Veteran populations. *We hypothesize that heavy tobacco exposure (>30 pack-years) will generate a disproportionately lower PFS and OS in Veterans compared to non-Veterans with HPV (+) OPC.*

Experimental approach.

Clinical datasets :1- VHA OPC cohort. Data from 8 VHA institutions contributing to the recently formed Head and Neck Cancer VA Consortium (VHA 1-VHA 8, **please see site key and letters of support**) will be aggregated pursuant to a detailed protocol submitted through the VA Central Institutional Review Board. PHI will be maintained behind the VA firewall, using a secure server at the primary institution and the existing Vista

Computerized Patient Record System (CPRS). The VA Inpatient and Outpatient MedSAS will be used to identify OPC patients in the study period (2010-2018) using International Classification of Diseases, 9th edition and 10th edition (ICD-9 and ICD-10) codes. Inclusion criteria for the cohort are: 1) pathologically confirmed diagnosis of OPC, 2) confirmed HPV status, and 3) curative intent treatment delivery. Exclusion criteria are: 1) previously treated head and neck squamous cell carcinoma, 2) distant metastasis at time of diagnosis, 3) palliative treatment selection. Demographic data, smoking status, tumor, and treatment characteristics will be collected from the medical record for all patients enrolled in this study. HPV status will be determined from the medical record by p16 immunohistochemistry status as outlined in the College of Anatomic Pathologists (CAP) guidelines or by direct HPV testing as performed during routine pathologic examination at the source institutions.⁴⁶ Oncologic outcomes included in the analysis will include locoregional control (LRC), distant metastasis (DM), progression free survival (PFS) and overall survival (OS). The VHA Vital Status File will be used to assess the time of death and OS for the subjects included in this study. 2- NCDB OPC cohort. The National Cancer Database (NCDB) will be used to generate 2010-2018 NCDB data for patients with HPV (+) OPC using the inclusion and exclusion criteria listed above. Over 15,000 HPV (+) OPC cases are available for the study time period which will facilitate the planned 2:1 matching planned below.^{47,48} Patients will be matched to the Veteran OPC cohort at a 2:1 ratio using age, T-classification, N-classification and subsite (tonsil vs. base of tongue vs. other OPC site).

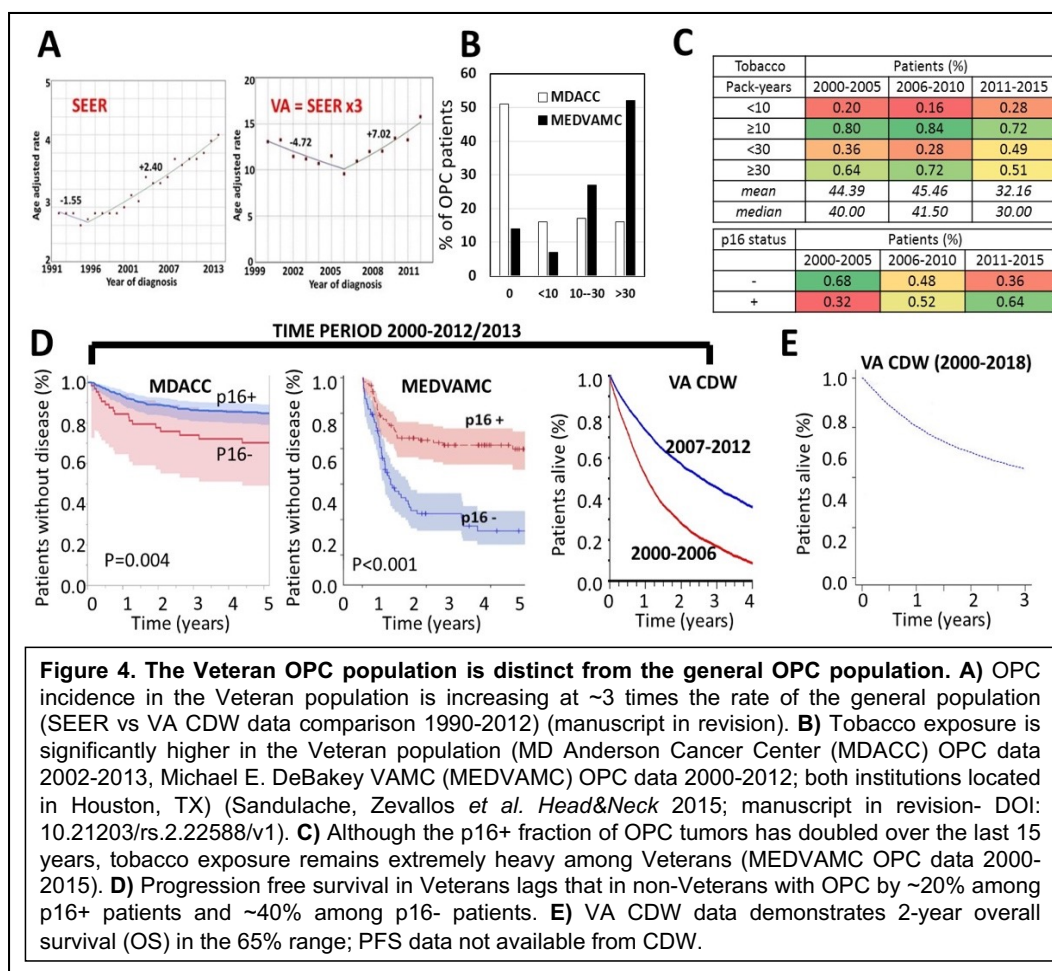
Statistical analysis. Univariate and multivariate analysis (Experiment 1.1) will be performed using standard statistical tools such as IBM SPSS Statistics (v25 or later). Univariate and multivariate tests will be considered statistically significant when the p-value is less than 0.05. Multivariate analysis will be conducted using clinical-pathologic variables with a p-value <0.1. The analysis detailed in Experiment 1.2 will be completed using a combination of SPSS and R, which is a free software package for statistical computing and analysis.⁴⁹⁻⁵¹ For a simple 2 proportion power analysis, with a differential LRC of 25% (85% for HPV+ non-smokers vs 60% for HPV+ smokers), a type I error rate of 5% and a power of 0.80, we will require only 150 events. Based on the LRC rate for our patient cohort, we will exceed the required number of events by > factor of 5. If we divide tobacco exposure into tertiles (0-30 pack years, 30-60 pack years and >60 pack-years) with a distribution of 40%, 40% and 20% respectively and relative PFS rates of 80%, 70% and 55% at 2 years, we should be able to generate sufficient power for the analysis detailed in Experiment 1.2. For Experiments 1.1 and 1.2, we anticipate that the sample size and follow-up that will be available for this study will allow detection of clinically important effects of tobacco with moderate to high power. For example, in one possible scenario that is consistent with preliminary data, we expect that low tobacco exposure (<30pk-yrs) will be associated with high (>80%) 2-year PFS rates, moderate exposure (30-60 pk-yrs) might be associated with lower (70%) rates, and high exposure (>60 pk-yrs) might be associated with relatively low (55%) rates. With about 1,500 cases distributed across exposure groups (40%, 40%, and 20% respectively), and assuming approximately exponential ‘time to recurrence’ with median follow-up of at least 3 years, we expect to have 99% power to detect an overall monotonic association (alpha=5%, two-tailed) in a Cox regression model. Linear contrasts between low exposure (reference group) vs intermediate, low vs high or intermediate vs high will be detectable with 70%, 99%, and 76% power, respectively. Power estimates are based on simulations with 5000 replications. Larger effects or more distinct thresholds will be detectable with greater power.

Experiment 2.1. Measuring the association of tobacco exposure on HPV (+) OPC with PFS and OS. We will measure the impact of tobacco exposure on PFS and OS via univariate and multivariate analysis using 2/3rds of the total patient number available for analysis. The remaining 1/3rd of patients will be held back as a validation cohort. Univariate analysis: Patients with HPV (+) OPC will undergo univariate analysis for PFS and OS using a Cox proportional hazards model and subsequent Wald test, with appropriate verification of assumptions. Individually tested variables will include: age, gender, race/ethnicity, T-stage, N-stage, M-stage, tobacco exposure (standard cut-offs) and treatment characteristics (external beam radiotherapy (EBRT) alone, EBRT + concurrent chemotherapy (EBRT+C) vs induction followed by EBRT (IC+EBRT)).

Multivariate analysis: All variables from the univariate analysis (UVA) with a p-value <0.1 will be included in the multivariate analysis (MVA). The MVA will be performed using a proportional hazard (Cox) regression model and subsequent Wald test. We have identified a cohort of 300 HPV (+) OPC patients treated at the participating institutions between 2010 and 2018. Incorporation of clinical data from this cohort of patients will allow us to evaluate the impact of HPV/p16 status on survival and generate relative hazard ratios for both tobacco exposure and HPV/p16 status as well as characterizing the interaction term between these two variables.

Experiment 2.2. Characterizing the type of effect for tobacco use on PFS and OS. We will conduct a ‘functional form’ analysis of the functional form of the tobacco exposure explanatory variable to elucidate the form (i.e. threshold, continuous-linear, continuous-nonlinear) of its relationship to risk of LR relapse. There are several approaches, the simplest of which examines a smoothed plot of residuals from the null survival model plotted against pack-years.⁵² Simple visual inspection of the data is often sufficient to determine whether the threshold effect represents a breakpoint or classification threshold with or without a transition component and then calculate the threshold or change point.^{49,53} An alternative is to model the functional form directionally using smoothing splines of the explanatory variables of interest. Analysis of deviance of nested models can be used to determine whether the additional complexity adds significance, with the goal of keeping things as simple as possible. The analysis of a potential threshold effect will be conducted using a combination of SPSS or SAS and R. N-fold cross-validation and/or bootstrapping will be used to assess performance. Finally, using a definitive Cox proportional hazards model, we will generate a nomogram for predicting 2 and 5-year PFS and OS for OPC patients using the above approach to modeling the impact of tobacco exposure.

Preliminary Data. We have demonstrated an increase in the incidence of HPV(+)OPC cases in the Veteran population.^{45,54} In a VHA OPC cohort, we previously demonstrated that >60% of patients had a >10 pack-year smoking history and that smoking was associated with worse oncologic outcomes.⁴⁵ Using national VHA administrative data of >12,000 OPC cases, we showed a statistically significant decline in OPC incidence between 2000 and 2006 (APC = -4.27, p <0.05), followed by a statistically significant increase between 2006 and 2012 (APC = 7.02, p <0.05) (**Figure 4A**). Compared to a non-VHA tertiary care center (MDACC), the rate of smoking in Veterans with OPC remains significantly higher (2000 to 2015) (**Figure 4B&C**) In Veterans, the rate of never-smokers ranged from 8 to 15% per year, compared to approximately 40% in recent OPC cohorts presented in the literature.^{1,10} PFS in Veterans lags that of a non-Veteran cohort with OPC by approximately 20% and 40% for HPV (+) and HPV (-) cases, respectively (**Figure 4D**); 2-year OS is significantly lower than that reported in civilian OPC cohorts (**Figure 4E**).^{1,10,47} These findings underscore the need for a Veteran-centric approach to understanding susceptibility to HPV (+) OPC, relationship to smoking status and overall oncologic outcomes as drivers of disease burden on individual patients and the VHA.



Expected results, pitfalls and alternative approaches. Completion of Aim 2 will allow us to estimate the interaction between HPV and tobacco exposure in Veterans, and to compare and contrast these parameters with those in the non-Veteran cohorts. As a point of clinical reference, the effect size of adding concurrent chemotherapy to radiation for OPC is estimated at approximately 7%.⁵⁵ The effect size for nodal metastasis without extranodal extension of approximately 20% and an additional effect size for extranodal extension of 15%.⁵⁶ Based on these comparators, an effect size for tobacco exposure in the 10-20% range would be considered highly clinically

significant making our current investigation clinically and translationally relevant. It is unclear whether tobacco exposure will impact PFS and OS in a continuous fashion or via a threshold effect. There exist examples in other pathologic conditions, including some linked to tobacco exposure (i.e. emphysema, lung cancer) for both types of interactions.⁵⁷⁻⁵⁹ Detection of a unique threshold would provide a critical first step in appropriate risk stratification for Veterans with OPC. However, as demonstrated by other investigators, even with the presence of a threshold, incremental decreases in PFS and OS are likely to be detected as a function of increasing exposure, with the exception of a breakpoint-based threshold/step which is highly unlikely from a biological perspective. As a result, we expect to generate a nomogram that will allow us to appropriately stratify Veterans with OPC based on tobacco exposure and other known variables. This is a unique deliverable that can be generated as part of this grant application and implemented widely throughout the VHA. Tobacco exposure is a patient reported variable, and subject to implicit, explicit and recall bias.⁶⁰⁻⁶² Nevertheless, current clinical decision making with regard to tobacco exposure uses patient reported data 10 pack-year threshold to define a “smoker” in the context of head and neck cancer.¹ Utilization of appropriate tobacco use measures has been shown to be effective in Veteran smokers, with the primary failure being an over-reporting of tobacco cessation, resulting in a potential under-estimation of tobacco exposure in the context of our planned analysis.⁶³ Alternatively, all patients undergo chest imaging in the form of either PET-CT or dedicated CT-chest as per institutional protocols. This imaging could be evaluated in a blinded fashion with relative scores assigned for the extent of lung emphysema as previously described,^{57-59,64} and be utilized as an alternative to patient reported pack-years in the Aim 1 analysis.

Aim 3: To project future HPV (+) OPC incidence and cancer-related mortality in the VHA. A Veteran-centric, evidence-based HPV (+) OPC simulation model will be constructed to describe disease progression in relation to

risk factors, diagnosis, treatments, and survival. We will incorporate HPV vaccination rates as well as smoking rates among HPV(+) OPC in order to address the impact of primary (vaccination) and secondary (smoking cessation) on the future incidence and cancer-related mortality associated with this disease. This model will be used to quantify disease burden in terms of HPV (+) OPC incidence, survival time, and quality-adjusted life years. We hypothesize that tailored HPV vaccination and smoking cessation efforts will reduce HPV (+) OPC incidence beginning in 2040, with a delayed improvement in survival and accompanying reduction in healthcare utilization and cost 3-5 years later.

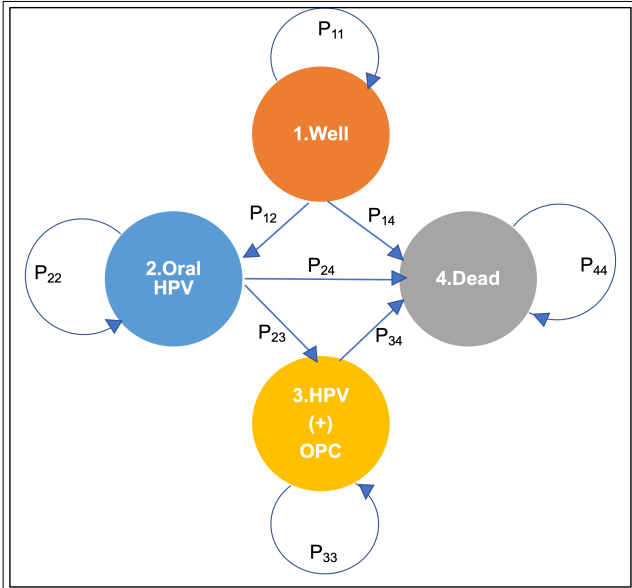


Figure 5. Markov Model Scheme. There are four health states included in the Markov Model: 1. Well (orange circle); 2. Oral HPV (blue); 3. HPV (+) OPC (yellow); and 4. Dead (grey). Each of blue arrows represent the probabilities of transitioning from one state to another.

Experimental Approach. Model scheme: We will construct a Markov model with a lifetime horizon to depict the natural history of HPV (+) OPC and its survival. The cycle length in this Markov model will be 5 years. In this Markov model, four health states will be included: 1) well (orange color in **Figure 5**); 2) oral HPV (blue); 3) HPV (+) OPC (yellow); and 4) dead (grey). The model starts with cycle 0 and cycles through cycle 1, 2 ... until termination (see termination condition in Monte Carlo simulation). In each cycle, an individual can either transition to another state or stay in the current state. For an individual in the well state, s/he will either transition to the oral HPV state with a probability (w.p.) P_{12} or the dead state w.p. P_{14} or stay in the well state for the next cycle w.p. P_{11} . If an individual is in the oral HPV state, then this individual will either transition to the HPV (+) OPC state w.p. P_{23} or the dead state w.p. P_{24} or stay in the oral HPV state w.p. P_{22} for the following cycle. However since there is evidence that HPV is controlled through immune surveillance, an individual cannot transition back to the well state.⁶⁵ If an individual is already in

the HPV (+) OPC state, then this individual will either transition to the dead state w.p. P_{34} or stay in the HPV (+) OPC state w.p. P_{33} for the following cycle. The dead state is an absorbing state, which means once an individual enters this state, s/he will not transition to another state. **Transition probabilities** (P_{ij} , $i, j = 1, 2, 3, 4$): Transition probabilities will be estimated from the data or obtained from published studies.^{1,4,47} Transition probabilities will be a function of an individual's demographics, e.g., age, sex, and characteristics, e.g., numbers of HPV vaccines and sexual partners, whether an individual has genital HPV infection, smoking status, alcohol consumption. No arrows between the states (**Figure 5**) if the transition probabilities=0. **Utilities** (U_j): Utilities are quality-of-life scores ranging from 0 (dead) to 1 (perfect health). They are used to discount life years to obtain quality-adjusted life years (QALYs).⁶⁶ Utility in a health state (U_j) can be computed as (1 minus disutility from that health state). We will use published data on sex-age-dependent utilities^{67,68} as the base utility for an individual. Utilities for an individual throughout the life course will be obtained by subtracting disutilities resulting from disease status from the base utility. Disutility for each health state will be obtained from published studies through the Cost-Effectiveness Analysis Registry database.⁶⁹

Model calibration and validation. Monte Carlo simulation: We will simulate a cohort of Veterans (n =number of unique Veterans, who utilized VA healthcare in 2008-2009 without a diagnosis of HPV (+) OPC) with demographics and characteristics drawn from the distributions of those of the Veteran population. The simulated cohort of Veterans will enter the Markov model. In each cycle, individual Veterans will draw a random number from the uniform distribution between 0 and 1 to determine whether s/he will transition to another health state in the next cycle, based on their transition probabilities. Transition probabilities will vary by time/Markov cycle/age,

because all of them are a function of age and other factors (see Table 1). The termination condition will be set when all individuals in the cohort enter the dead state. Half-cycle correction will be used to account for transitions occur at any point during the cycle. When the model is terminated, the life histories of individual veterans will be summarized and aggregated in terms of number of incidental HPV (+) OPC in each year and survival for patients diagnosed with HPV (+) OPC. Calibration and validation: We will use odd years data of incidence of HPV (+) OPC after 2010 and median survival at the time of study for Veterans who utilized VA healthcare in 2008-2009 to calibrate the transition probabilities in the model. Model calibration will be performed and the parameters will be adjusted using the method proposed in Taylor et al. (2011): the weighted mean percentage deviation from the predicted values (the aforementioned model outputs) to the observed values will be minimized using the Nelder-Mead algorithm.⁷⁰ We will use even years data of incidence of HPV (+) OPC after 2010 and median survival at the time of the study to validate the model. Graphs of predicted and actual values will be plotted for visual examination. In an unlikely case in which the validation fails, the model will be carefully examined and recalibrated, followed by revalidation to ensure the reliability of the model until the model successfully validates.

Simulations Simulation under the status quo: Using the calibrated and validated model, Monte Carlo simulations will be performed on a cohort of Veterans (n =number of unique Veterans, who utilized VA healthcare in 2020 without a diagnosis of HPV (+) OPC) using the same way described above. The model outputs will be summarized and aggregated in terms of numbers of incident oral HPV and HPV (+) OPC in each future calendar year and survival years and QALYs for patients diagnosed with HPV (+) OPC. Standard errors will also be computed from the outputs of the Monte Carlo simulation. Simulations under Scenario 1: uptake of HPV vaccination: We will use a) the trend of the prevalence of HPV vaccination to predict the future trend; b) a hypothetical range of uptake of HPV vaccination in the population age <45. Monte-Carlo simulation will be performed on the simulated 2020 Veteran population with the assumption of the uptake of HPV vaccination. The uptake of HPV vaccination will decrease the transition probability, P_{12} , probability transitioning from well to oral HPV (**Figure 5**), thus reduce the numbers of incidental oral HPV infection and HPV (+) OPC, and increase life expectancy and quality-adjust life expectancy of the population. Simulations under Scenario 2: smoking cessation: We will use a hypothetical range of the reduction in the prevalence of current smokers in the simulated population. Monte-Carlo simulation will be performed on the simulated 2020 Veteran population with the assumption of reduction in the smoking prevalence through effective smoking cessation interventions. Smoking cessation will reduce the transition probabilities from a better health state to a worse health state, i.e., P_{12} , P_{14} , P_{23} , P_{24} , and P_{34} . We expect that the numbers of incidental oral HPV infection and HPV (+) OPC will decrease, and life expectancy and quality-adjust life expectancy of the population will significantly increase.

Sensitivity Analyses: One-way sensitivity analyses using a reasonable range of values for each of the transition probabilities and utilities will be conducted. A tornado diagram will be generated to evaluate which variable(s) are sensitive the interested model outputs. We will explore the variation in the uptake of HPV vaccination in terms of complete vs partial HPV vaccinations and age of vaccination. We will also explore variation in the effectiveness of smoking cessation interventions, e.g., % quit smoking immediately, % quit in 5 years. The impact of these variations on the model outputs will be evaluated. We will also perform threshold analyses to identify the minimum level of uptake of HPV vaccination or smoking cessation to reach the policy target.

Summary and future directions. HPV (+) OPC cancer poses a clear and critical danger to Veterans. The rapid increase in disease burden, coupled with increasing costs associated with delivery of effective, definitive treatment for OPC generate a substantial, yet unpredictable burden to the DOD and the VHA. The current proposal will generate a predictive model able to quantify both individual patient risk and overall disease burden at a healthcare system levels. This is a critical first step in developing targeted interventions designed to ameliorate disease burden using targeted interventions aimed at HPV vaccination and smoking cessation. If we are successful, we

expect fairly rapid implementation through existing DOD and VHA healthcare delivery and prevention efforts across the country.

SAMPLE RECRUITMENT: The proposed project is a clinical database study and will not require sample recruitment.

IMPLEMENTATION AND DISSEMINATION PLAN

The current proposal presents an opportunity to save Veterans lives and limit morbidity through improved understanding of opportunities to introduce prevention strategies through HPV vaccination. The rapid increase in disease burden, coupled with increasing costs associated with delivery of effective treatment for OPC generates a substantial but unpredictable burden to the VHA. The current proposal will generate a predictive model able to quantify both individual patient risk and overall disease burden at a healthcare system levels. This is a critical first step in developing targeted interventions designed to ameliorate disease burden using targeted interventions aimed at HPV vaccination and smoking cessation. If we are successful, we expect fairly rapid implementation through existing VHA healthcare delivery and prevention efforts across the country. This study will focus exclusively on Veterans, specifically targeting veterans less than 45 year of age eligible for HPV-vaccination who are retiring from active-duty military and entering the VHA system.

VA National and Program Office Priorities: This study directly supports VA national priorities in improving access to care, improving health disparities, and promoting population health. The results of this study have the potential to directly address VHA priorities by promoting opportunities for head and neck cancer prevention through vaccination. This grant also address current and future health disparities among Veterans through the indentification of avenues to provide more equitable access to preventative care, including HPV vaccination and smoking cessation. Finally, this project will determine the impact of smoking in current HPV(+) OPC outcomes. This will lead to a more Veteran-centric approach to treatment for this disease, and may lead to VA-centric clinical trials for this disease. The VISN system would be the ideal operational partner to “own” the results of this study since we will be able to provide region-specific data on HPV vaccination uptake, smoking, and barriers to care. This would allow VISN leadership to address the specific needs of Veterans in their VISN when designing and implementing interventions to prevent HPV(+) OPC.

Long Term Usage and Goals: We intend to share the results of our study as widely as possible with both the scientific community and the public. We plan to share the results of this project in several ways to make it available to the community of scientists and the public interested in HPV (+) OPC epidemiology and outcomes. This effort will minimize duplication of research efforts and encourage collaboration with other investigators. Besides the obvious dissemination of data through publications, the PIs and members of their laboratories usually attend the annual scientific meetings where work generated from this proposal would be presented. They also usually give 1-2 invited seminars/year at other institutions and National and International meetings. Data from this study will also be shared with the VHA OPC Working Group. Finally, we will provide ready access of data and results VHA and VISN leadership so that these data could be used to develop cancer treatment and prevention strategies. Each Aim of this proposal incorporates short and long-term goals and deliverables that will have a direct impact on improving the lives of Veterans:

Aim 1: Short-term: Upon completion of Aim 1, we will define for the first time the current prevalence of HPV vaccination among eligible Veterans (<45 years of age), and define demographic and socioeconomic barriers to HPV vaccination. These data could serve as a baseline to assess the impact of future targeted interventions. Additionally, these data will be shared through publications and with VHA leadership in order to design targeted interventions. Regional reports will be provided by VISN to allow for tailored HPV vaccination profile. *Long-*

term: The VHA may also consider adding automated clinical reminders for HPV-vaccination into the EMR system based on findings from this study and sustained VISN focused cancer prevention interventions.

Aim 2: Short-term: Delivery of a Veteran-specific nomogram to improve risk stratification for Veterans with OPC. This will be provided to the VHA community through publication and distribution to clinicians within 1 year of completion of the grant. *Long-term:* Understanding the current state of HPV (+) OPC in the VHA will allow for clinical trials focused on high-risk populations, including in smokers. It will also address the potential role for treatment de-intensification in the VHA population.

Aim 3: Short-term: A novel HPV (+) OPC projection model would allow investigators and VHA leadership to project the future burden of HPV (+) OPC in Veterans over the next several decades, including cancer incidence, survival, and impact on the quality of life in Veterans. *Long-term:* Public health investigators will be able to estimate how vaccination and smoking cessation would impact the future incidence and outcomes of HPV (+) OPC in Veterans. These data could be used to justify HPV vaccination efforts both in Veterans as well as active-duty military personnel. These are our most powerful tools against this deadly cancer and if implemented strategically and in a data-driven way could lower suffering and death from OPC among Veterans in the future. Prevention of HPV (+) OPC would have a profound impact on the VHA population for decades to come.

Future Directions: While this study is focused on HPV-vaccination, it has the potential to be the first step towards developing a tool to monitor vaccination programs among Veterans and identify populations at risk for under-vaccination. This could be done as a “Veteran Immunization Data Analytics (VIDA)” tool designed to provide aggregate, de-identified data on trends in vaccination uptake and barriers. This tool could be used to monitor current and future vaccination campaigns. It would draw data from VISTA and CPRS in order to provide a complete picture of vaccination and undervaccination in the VHA population. The impact of this research will be measured through improved HPV vaccination rates among eligible Veterans, and ultimately a reduction in HPV(+) OPC in the VHA population. The impact of this work will also be measured through adoption of our HPV(+) OPC prediction model at the VISN and VHA level in guiding policies regarding HPV vaccination.

PROJECT MANAGEMENT PLAN

Project Management Team																
Table 2. Research Timeline		(Quarters: June 2023-May 2027)														
Quarter	1--1	1--2	1--3	1--4	2--1	2--2	2--3	2--4	3--1	3--2	3--3	3--4	4--1	4--2	4--3	4--4
IRB approval																
Aim 1. HPV-Vaccination Rates/Barriers																
Experiment 1A: Development of Administrative Dataset																
Experiment 1B: Demographic/Regional Barriers																
Aim 2. HPV(+) OPC Outcomes and Smoking in the VHA																
Experiment 2A: Association of Smoking Status and Survival																
Experiment 2B: Effect Measure of Smoking on Outcomes																
Aim 3. Simulation Modeling of HPV(+) OPC in the VHA																
Manuscript Preparation																
Results Dissemination to VHA VISN Leadership																

Research Team and Roles: The investigative team assembled for the proposed study has a proven record of success in their research areas with complementary expertise. Dr. Zevallos and Dr. Sandulache have been leaders in understanding head and neck cancer epidemiology and clinical outcomes in the VHA population and have worked together for the last decade. Dr. Mazul is an expert in head and neck cancer disparities, and has published extensively with Dr. Zevallos on this topic. Dr. Chang is a nationally-recognized health services researcher who has worked extensively with VHA administrative data and the develop of simulation models. Collectively, the team has collaborated on a number of projects on HPV (+) oropharyngeal cancer specific to the Veteran population including extensive experience working VHA administrative data.⁷¹⁻⁷⁶

Jose P. Zevallos, MD, MPH (Principal Investigator, no salary support requested): Dr. Zevallos is the Division Chief of Head and Neck Surgical Oncology and an Associate Professor in the Department of Otolaryngology/Head and Neck Surgery at Washington University School of Medicine. Dr. Zevallos is also a staff surgeon at the John Cochran VA Medical Center in St. Louis. He also holds a position as adjunct associate professor of Epidemiology in the UNC Gillings School of Global Public Health, and is a member of the Siteman Cancer Center. Dr. Zevallos is currently the PI of an NCI-funded R01 examining the role of gene expression subtype on treatment response in head and neck cancer. He is also PI of the Head and Neck Cancer Tissue Acquisition Protocol at Washington University School of Medicine/Siteman Cancer Center. He has previously received funding from the Department of Veterans Affairs VISN16 Pilot Project Grant to Study the Epidemiology of Oropharyngeal Cancer in Veterans. **Role:** As the Principal Investigator on this project, Dr. Zevallos will provide guidance on overall strategy, study design, patient cohort selection, clinical indications to be addressed and evaluation of clinical utility. He will also serve as the direct mentor for the Washington University post-doctoral researcher collaborating on this study and he will lead research regular team meetings.

Vlad C. Sandulache MD PhD (Co-Investigator, no salary support requested): Dr. Sandulache is an Assistant Professor in the Bobby R. Alford Department of Otolaryngology Head and Neck Surgery at Baylor College of Medicine and a Staff Physician in the Otolaryngology Head and Neck Surgery Section of the Michael E. DeBakey Veterans Affairs Medical Center Operative Care Line. He has significant experience examining both the unique clinical outcomes of active and former service members with oropharyngeal squamous cell carcinoma (SCC), as well as an active laboratory devoted to identifying signatures unique to smoking-related oropharyngeal SCC. He will assist in all aims, with a focus on Aim 2 that examines HPV (+) oropharyngeal cancer incidence and outcomes on a national at the VHA. Specifically, Dr. Sandulache will dedicate his effort to providing clinical insight into this disease process at the VHA and guidance on study approach, design, and analysis. **Role:** Dr. Sandulache will work closely with Dr. Zevallos and the other members of the team to complete the planned analysis as well as manuscript preparation and review. His experience at the VHA and working with veterans will also help inform Veteran engagement strategies, such as convening a panel of veterans at high-risk of HPV (+) OPC to obtain perspectives on HPV-vaccination, following completion of Aim 1 of the study.

Su-Hsin Chang, PhD SM (IPA Co-Investigator, 15%): Dr. Chang is Associate Professor in the Division of Public Health Sciences, Department of Surgery, Washington University School of Medicine (WUSM). Dr. Chang is an applied econometrician and a health services researcher by training. Dr. Chang's research has focused on health and economic burden of obesity and intervention of obesity, using large-scale data analyses and simulation modeling. Dr. Chang's research was funded by AHRQ and NIH (K01 HS022330, R21 DK110530, R01 CA253475, and U01 CA265735). Dr. Chang currently leads several ongoing IRB-approved VA studies. She has been a Without Compensation (WOC) Veterans Affairs (VA) investigator since 2013. She has 9 years of experience in utilizing the VHA databases. Since 2013, Dr. Chang teaches semester-long "Decision Analysis for Clinical Investigation and Economic Evaluation" every fall semester in the Master of Population Health Sciences degree program at the MUSM. Through this class, Dr. Chang mentored the students on several simulation studies (PMID 25552420, 28212973, 26319488, 29428700). **Role:** Dr. Chang will work with the team to conduct the proposed data analyses and simulations for Aim 1-3. This includes putting together Data Access Request Tracker (DART) applications for data request and overseeing data merging, abstraction, cleaning, management, and analyses using data from various datasets in the VHA database. Moreover, she will work with the team to write and review manuscripts.

Angela L. Mazul, PhD, MPH (IPA, Co-investigator, 5% effort): Dr. Mazul is a head and neck cancer epidemiologist and Assistant Professor in the Department of Otolaryngology/Head and Neck Surgery at Washington University School of Medicine. She holds a joint appointment in the Division of Public Health

Sciences, Department of Surgery. Dr. Mazul is an expert in head and neck cancer epidemiology and cancer disparities, with a specific emphasis on HPV (+) oropharyngeal squamous cell carcinoma. Dr. Mazul currently studies head and neck cancer disparities and is funded through a K01 grant from the National Institutes of Health/National Institute of Minority Health and Health Disparities (NIMHD) **Role:** Dr. Mazul will assist in analysis of healthcare disparities in the context of HPV vaccination.

To be named Post-Doctoral Research Associate (IPA, 100% effort): The post-doctoral associate will be responsible for working under the guidance of Dr. Zevallos on study design and methodology. This researcher will help apply for IRB approval of the proposed study and will help with the process of conceptualizing study design and approach as well as authoring manuscripts.

To be named Clinical Research Coordinator (100% effort): A research coordinator will be recruited from the VHA to oversee administrative aspects vital to ensuring the study can be initiated and sustained throughout the funding period. **Role:** This person will be familiar with the on-boarding process and research training requirements of the VHA to ensure that all personnel in the study are up to date with their paperwork and maintain eligibility to work with VHA data.

To be named Data Analyst (100% effort): A data analyst with an extensive background in biostatistics or epidemiology, and experience working with VHA datasets and the VINCI environment will maintain the IRB-approvals and request data. Additional responsibilities will include working with Dr. Chang and a VA decision modeler to conduct the proposed research and perform data request and analyses, model building, and simulations.

To be named Decision Modeler (100% effort): A decision modeler with a background in decision science, simulation modeling, or engineering, will be recruited from the VA to work on this study. **Role:** The decision modeler will help build the Markov models for the natural history of OPC using both TreeAge and Matlab. S/he will perform a thorough search and review on the relevant topics on Pubmed and tabulate the results for Dr. Chang's review to identify the gap in the literature for the specific parameters needed in the model. If necessary, a systematic review and meta-analyses will be performed.